10μm pitch family of InSb and XBn detectors for MWIR imaging

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ABSTRACT

There has been a growing demand over the past few years for infrared detectors with a smaller pixel dimension. On the one hand, this trend of pixel shrinkage enables the overall size of a given Focal Plan Array (FPA) to be reduced, allowing the production of more compact, lower power, and lower cost electro-optical (EO) systems. On the other hand, it enables a higher image resolution for a given FPA area, which is especially suitable in infrared systems with a large format that are used with a wide Field of View (FOV). In response to these market trends SCD has developed the Blackbird family of 10 μm pitch MWIR digital infrared detectors. The Blackbird family is based on three different Read-Out Integrated Circuit (ROIC) formats: 1920×1536, 1280×1024 and 640×512, which exploit advanced and mature 0.18 μm CMOS technology and exhibit high functionality with relatively low power consumption.

Two types of 10 μm pixel sensing arrays are supported. The first is an InSb photodiode array based on SCD's mature planar implanted *p-n* junction technology, which covers the full MWIR band, and is designed to operate at 77K. The second type of sensing array covers the blue part of the MWIR band and uses the patented XBn-InAsSb barrier detector technology that provides electro-optical performance equivalent to planar InSb but at operating temperatures as high as 150 K. The XBn detector is therefore ideal for low Size, Weight and Power (SWaP) applications. Both sensing arrays, InSb and XBn, are Flip-chip bonded to the ROICs and assembled into custom designed Dewars that can withstand harsh environmental conditions while minimizing the detector heat load. A dedicated proximity electronics board provides power supplies and timing to the ROIC and enables communication and video output to the system. Together with a wide range of cryogenic coolers, a high flexibility of housing designs and various modes of operation, the Blackbird family of detectors presents solutions for EO systems which cover both the very high-end and the low SWaP types of application. In this work we present in detail the EO performance of the Blackbird detector family.

Keywords: Infrared Detector, InSb, XBn, Focal Plane Array, 10 μm pixel, High Operating Temperature.

1. INTRODUCTION

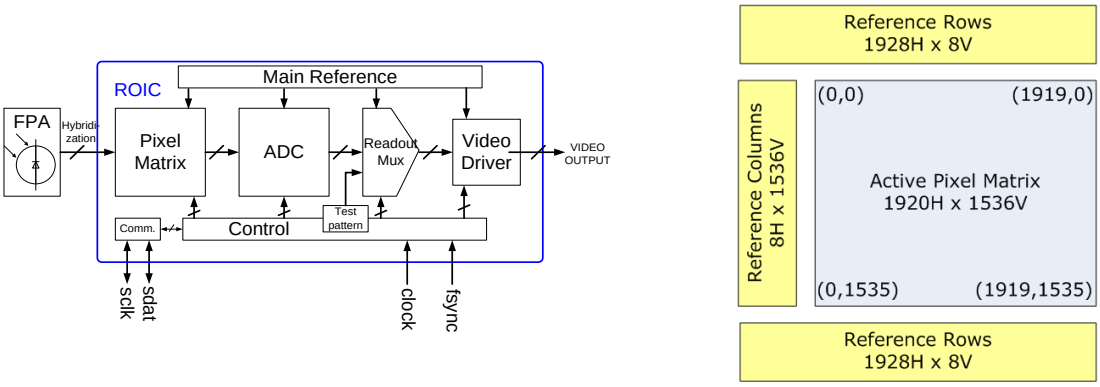
The motivation for reducing the pixel size in an array detector is clear from the system point of view[1]. Modern electro-optical (EO) systems are designed with a more compact, lower power, and lower cost solution in mind than their larger predecessors. For example, a pixel size reduction at constant Instantaneous Field of View (IFOV) enables a corresponding reduction in the size of the system optics, since the focal length is decreased. For a given Focal Plane Array (FPA) format, the detector area decreases with pixel size, reducing the Dewar dimensions and thus the required cooling power. It is therefore possible to reduce the overall Size, Weight and Power (SWaP) of the entire system. Alternatively, if the system size can stay constant the FPA format can be increased, which improves the image resolution and therefore performance metrics such as the range for target Detection, Recognition and Identification (DRI). The Blackbird family of detectors has a small pixel size of 10 μm. The smaller pixel meets the latest requirements for a wide spectrum of applications, from very high-end to very low SWaP Mid-Wave Infrared (MWIR) systems.

In the past, 2D second generation IR detectors were mostly used for imaging. Nowadays, larger format detectors that combine a large Field of View (FOV) with high resolution, enable applications such as Persistent Surveillance, Situational Awareness, Infrared Search and Track, Missile Warning Systems and numerous other EO systems. IR detectors with a 15μm pixel such as SCD's Pelican-D and Hercules, with VGA and SXGA formats respectively, have been widely used in these systems. The Blackbird 1280×1024 (SXGA) Integrated Detector Cooler Assembly (IDCA) has similar dimensions and mechanical interfacing to Pelican-D and thus enables the retrofitting of VGA systems to XGA or SXGA formats, providing a significant improvement of resolution simply by replacing the detector and in some cases reducing the optics F#. Alternatively, a reduction of the system size is possible by replacing the SXGA Hercules detector by the SXGA Blackbird detector. Another alternative is to upgrade a system based on the SXGA Hercules by the Blackbird 1920×1536 detector, which has the same dimensions and mechanical interfacing, enabling increased performance with high resolution and high frame rate.

The Blackbird family can incorporate InSb photodiodes based on SCD's standard implanted *p*-on-*n* technology that covers the full MWIR atmospheric window and operate at 77 K. Alternatively, a patented XBn-InAsSb barrier detector technology[2] can be used[3], which covers the highly transparent blue part of the MWIR window and which is a truly High Operating Temperature (HOT) technology operating up to ~150 K. Increasing the FPA operating temperature is extremely important for further reducing the cooling demands of the detector with a major impact on the cooler performance and hence on the SWaP of the IDCA and system. Improved reliability and Mean Time to Failure (MTTF) is an additional benefit of high operating temperatures, and is a crucial factor in 24/7 Persistent Surveillance, Terrain Dominance, Homeland Security, and Border Control systems.

2. ROIC

Since many years SCD has been developing and manufacturing Readout Integrated Circuits (ROICs) with Analog to Digital Conversion (ADC) at the focal plane, implemented using advanced 0.18 µm mixed-signal CMOS process. This feature has been incorporated in several detectors such as Sebastian, Pelican-D, and Hercules[4-5]. The main ROIC building blocks are matrix readout circuits, ADCs, video readout multiplexer (MUX), and output driver. A schematic functional block diagram of the ROIC is presented in figure 1(a). The ADCs are integrated on chip at opposite sides of the active matrix in two rows of 1920 / 1280 / 640 column ADCs including an output MUX which multiplexes the ADCs to the chip output[6]. The MUX is compact enough not to increase the die size significantly, despite the 10 µm column pitch. Reading the pixel signals is possible from two column ADCs simultaneously, using a high speed digital 13 bit sub-LVDS video interface developed to operate at the required data bandwidth, while using a reasonable pin-count. In Table 1 we show that the Blackbird ROICs work at high frame rates with relatively low power. A single ADC channel column is based on a dedicated Dual Ramp convertor designed for low noise and low power consumption reaching less than 160 µV input-referenced noise and consuming less than 35 µW power, while reaching a 95 kHz sampling rate.

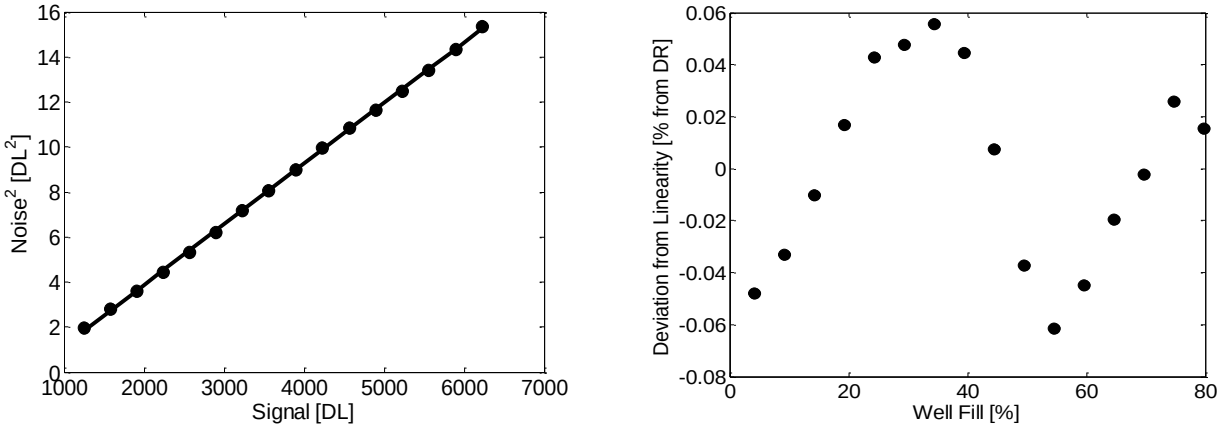


(a) (b)

Fig. 1 (a) ROIC functional block diagram. (b) A scheme of the pixel matrix structure in Blackbird-1920

Figure 2(a) presents the square of the noise measured as a function of the signal in an InSb Blackbird-1920 FPA at 77 K, for different integration times. A linear dependence is observed which indicates that the detector is shot-noise limited. These results represent a fine readout process from the ROIC pixel which does not introduce any additional noise components such as 1/f, Random Telegraph Signal (RTS), or those related to various leakage mechanisms which tend to increase at lower temperatures. The deviation from linearity of the ROIC and FPA is presented in Figure 2(b) and shows low non-linearity of less than 0.06% of the full Dynamic Range (DR) from 5% to 80% capacitor well-fill. Another building block of the Blackbird ROIC is the matrix readout circuit (figure 1 (b)). This is carefully designed to provide an optimal peripheral electronic environment for operation of the 10×10 µm^2 InSb or XBn, *p*-on-*n* polarity photovoltaic pixels, which operate at 77 and 150 K respectively. It allows a wide range of controllable biasing and gain for different applications and scenarios. Several conversion gain options are implemented at the pixel level to enable selection of the

most suitable gain mode. At null integration time, no photoelectrons are collected in the integration capacitors, so the readout noise is then a property of the ROIC, alone. This readout noise is then dominated by Johnson-Nyquist noise and hence is temperature dependent.



(a) (b)

Fig. 2 (a) Squared noise as a function of signal in digital levels measured for the Blackbird-1920 InSb FPA, for increasing integration times. (b) Deviation from linearity, where the well-fill is varied using the integration time.

The low noise characteristics of the various Blackbird ROICs together with other key parameters are summarized in Table 1.

Parameter	**Blackbird 1920**	**Blackbird 1280**	**Blackbird 640**
Format	1920×1536	1280×1024	640×512
Pixel Size	10×10 μm^2		
Output	Digital 13 bit		
Well Fill Capacity / Readout Noise at 77K *	0.3Me$^-$ / 60e$^-$ 0.5Me$^-$ / 90e$^-$ 2.0Me$^-$ / 330e$^-$ (260e$^-$ ITR) 4.0Me$^-$ / 800e$^-$		0.7Me$^-$ / 150e$^-$ 2.0Me$^-$ / 330e$^-$ (260e$^-$ ITR)
Integration modes	ITR, IWR		
Frame cycle control	Free running, System control		
Maximum Frame Rate	120 Hz (4 video ports) 60 Hz (2 video ports) 30 Hz (1 video port)	180 Hz (2 video ports) 90 Hz (1 video port)	350 Hz (1 LVDS ports) 180 Hz (1 LVCMOS ports)
FPA power consumption	400 mW at 120 Hz 210 mW at 60 Hz 110 mW at 30 Hz	210 mW at 180 Hz 110 mW at 90 Hz 85/50^{cmos} mW at 30 Hz	210 mW at 350 Hz 110 mW at 180 Hz 85/50^{cmos} mW at 60 Hz
Maximum Frame Rate in 2×2 Binning mode	450 Hz	600 Hz	1000 Hz
* For 150K operating temperature, the Readout Noise may increase by 30% at most.			

Table 1 Key features of the three Blackbird ROICs

For the Blackbird family SCD has adopted a new approach to increase the ROIC functionality, autonomy and compatibility with standard system video and serial communication protocols. As a result Blackbird can be integrated easily into a system without any additional electronics boards. The ROIC enables windowing in the horizontal direction and flipping of the horizontal and vertical readout directions. Windowing at the readout level increases the maximal frame rate for a given operational configuration. In addition, a 2×2 pixel binning feature implemented at the ROIC level improves the Signal to Noise Ratio (SNR) and considerably increases the maximum frame rate, with an effective pixel size of 20 μm.

The main features of the ROIC are:

- 10μm pitch,
- Multiple in-pixel gain modes
- Simple interfacing and operation
- Self-initialization from external E^2PROM
- Internal frame sequence
- Glue-less video output
- Standard serial interface (E^2PROM compatible)
- High frame rate
- Reduced bandwidth & power consumption modes
- Reset options: software command reset, Hard reset
- Temperature reading by video output
- Direct-access temperature diode, available also when the ROIC is off

In summary, the Blackbird family of ROICs exhibit very low noise and highly linear performance enabled by the innovative ADC channels[7-8], together with relatively low power. The controllable in-pixel gain and bias, together with the additional functionality at the readout level, the video output options and the possibility of integrating both InSb or XBn devices with a temperature range from 77 to 150 K, are all part of what makes the Blackbird ROIC versatile for various IR detectors used in a wide range of EO systems.

3. INSB PHOTO-DIODE ARRAY

The Blackbird InSb sensing array of 10×10 μm^2 pixels is based on SCD's mature InSb planar, implanted *p-n* diode technology, which is in production over many years for various formats and pitches. The 10 μm pixel in the Blackbird detector is based of the well-used 15 μm pixel[5]. The successful scale down to a 10 μm pitch was achieved in spite of the increased ratio of surface to volume that imposes new design rules necessary to maintain key parameters such as low Dark Current (I_{dark}) and high External Quantum Efficiency (QE). Over the last few years, SCD has introduced 10 μm InSb pixel arrays with three different formats: 1920×1536, 1280×1024, and 640×512. These pixel arrays are integrated with the Blackbird family of ROICs using Flip-chip indium bump technology with reduced bump dimensions. The resulting FPA exhibits high performance with QE >80% and I_{dark} <0.8 pA at 77 K. In Figure 3(a) we present a histogram of the dark current values for all of the InSb diodes in a single 1920×1536 Blackbird FPA. The narrow distribution (~0.2 pA) indicates a high level of dark current uniformity over the array, as can also be seen in the dark current map in Figure 3(b). As shown in Figure 3(c), the reproducibility of the InSb diode fabrication process is demonstrated through the

narrow statistical spread of the average dark current values (for all pixels in the FPA) in 70 FPAs from SCD's Blackbird production line.

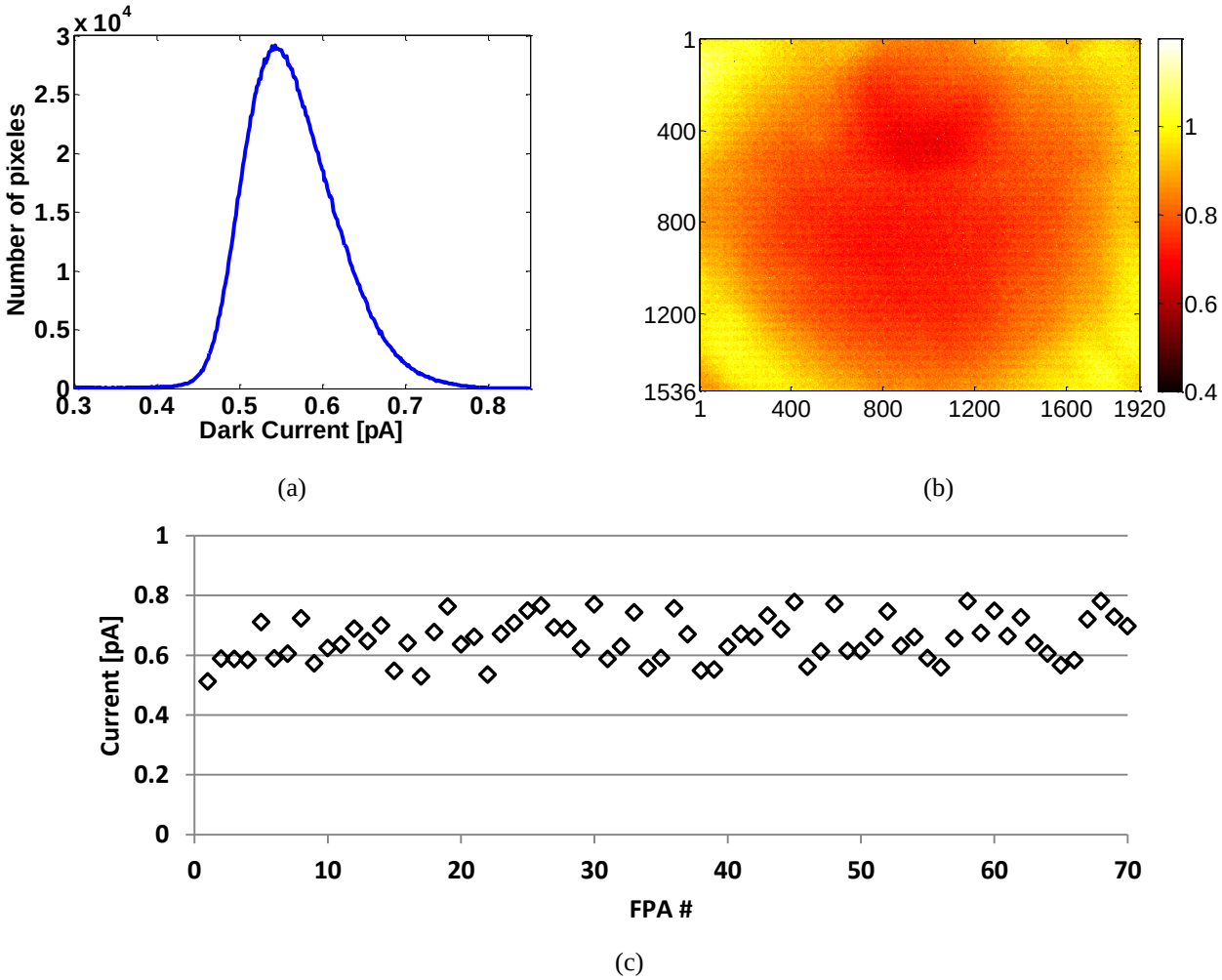
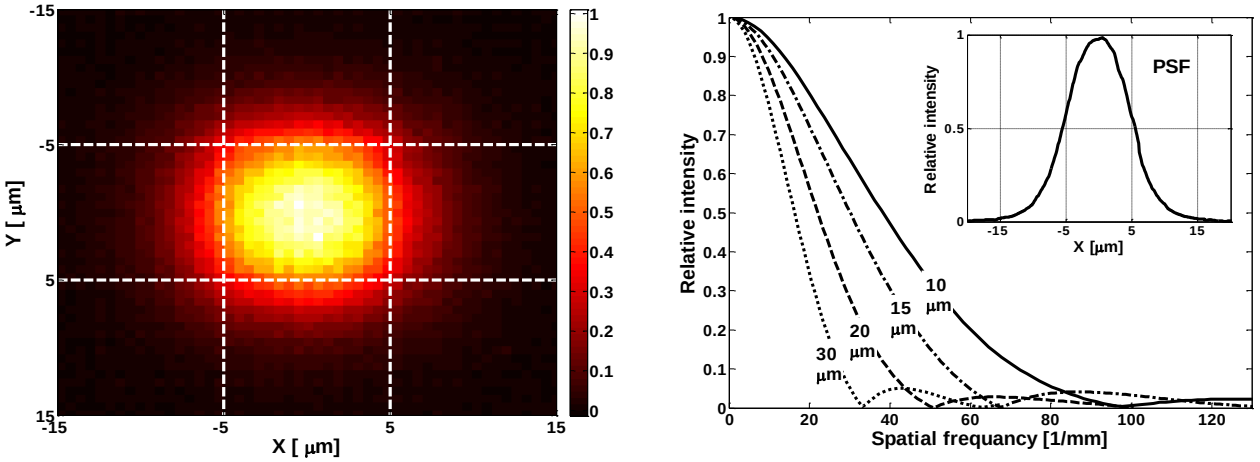


Fig. 3 Dark current (in pico-Amperes) measured in InSb Blackbird-1920 FPA at 77 K and plotted as (a) pixel distribution in the array, (b) 2D map, (c) mean values of 70 FPAs from SCD's production line

Another important issue in FPA image quality is the inter-pixel cross-talk (XT). This is the fraction of the light signal falling on a given pixel that is detected by one of its neighbors. The Blackbird FPA exhibits low cross talk characteristics where 56% of the total light signal falling on the entire pixel area is detected in the same pixel, 9% is detected in each of the four nearest neighbors, and 2% in each nearest–diagonal neighbor. The remaining 2% of the light signal is detected at the next line of nearby pixels. The cross-talk is most conveniently quantified by the Point Spread Function (PSF), a measurement of which is presented in figure 4(a) and in the inset of figure 4(b). The Modulation Transfer Function (MTF) is the Fourier transform of the PSF and is shown in figure 4(b), for different detector pitch values down to 10 μm. The MTF represents the amplitude of a spatially periodic signal detected by the FPA as a function of the spatial frequency. The PSF is measured using a gold mask with 400 square openings 2.5×2.5 μm^2 in area on the back side of the

FPA, where each opening is shifted by a different distance from the pixel center. The PSF can also be measured using a knife edge technique. Both techniques give a similar value of MTF = 0.4 at half the Nyquist frequency.



(a) (b)

Fig. 4 (a) Measured PSF image of InSb 10µm pitch detector, where the pixel borders are indicated by white dashed lines (b) MTF curves of InSb pixels in FPA with four different pitches: 30, 20, 15 and 10 µm, corresponding to SCD's Blue Fairy, Sebastian, Pelican and Blackbird FPA's, respectively. Insert: Cross section of the PSF image (Fig. 4a) at Y=0.

4. XBN BARRIER DETECTOR ARRAY

For the last few years SCD has been producing 640×512 and 1280×1024 MWIR detectors with 15 µm pitch based on the novel XBn-InAsSb barrier detector technology. This technology has outstanding electro-optical performance at operating temperatures as high as 150 K[9,10,11], however, with a cut-off wavelength of 4.2 µm covering only the "blue" part of the MWIR spectrum. Following the trend of pixel shrinkage as for the InSb pixel, and with the availability of the Blackbird family of ROICs designed to perform also at a high operating temperature, SCD has developed three new XBn arrays with a 10 µm pitch and formats of 1920×1536, 1280×1024 and 640×512. As already mentioned above, key parameters that determine the performance of the detector from the device point of view are quantum efficiency (QE) and dark current. High QE and low dark current yield a better SNR, and homogeneous QE and dark current implies good signal uniformity. Usually the main contribution to the dark current comes from a "Generation - Recombination (G-R) current" that is larger than the "Diffusion current" by several orders of magnitude and has a strong dependence on the FPA temperature. XBn devices are designed to have no depletion zone in the narrow band gap sensing material, and hence the G-R current is essentially totally suppressed. This leaves the much smaller Diffusion current coming from the narrow bandgap photon absorbing region, as the dominant source of dark current. In this way it is possible to elevate the operating temperature to 150 K with a typical dark current of only 0.2 pA, whose distribution in a typical 1280×1024 XBn Blackbird FPA is shown in figure 5(a). This width of the distribution is very narrow, corresponding to a dark current that essentially varies by ~10% across all pixels. Figure 5(b) shows the dark current map of the FPA

demonstrating very good spatial uniformity. The good reproducibility of the XBn fabrication process can be seen in figure 5(c), where a narrow statistical spread exists for the average dark current in 65 FPAs from SCD's Blackbird XBn production line.

The measured QE in XBn FPA is typically 70%, and the MTF is 0.36 at half the Nyquist frequency.

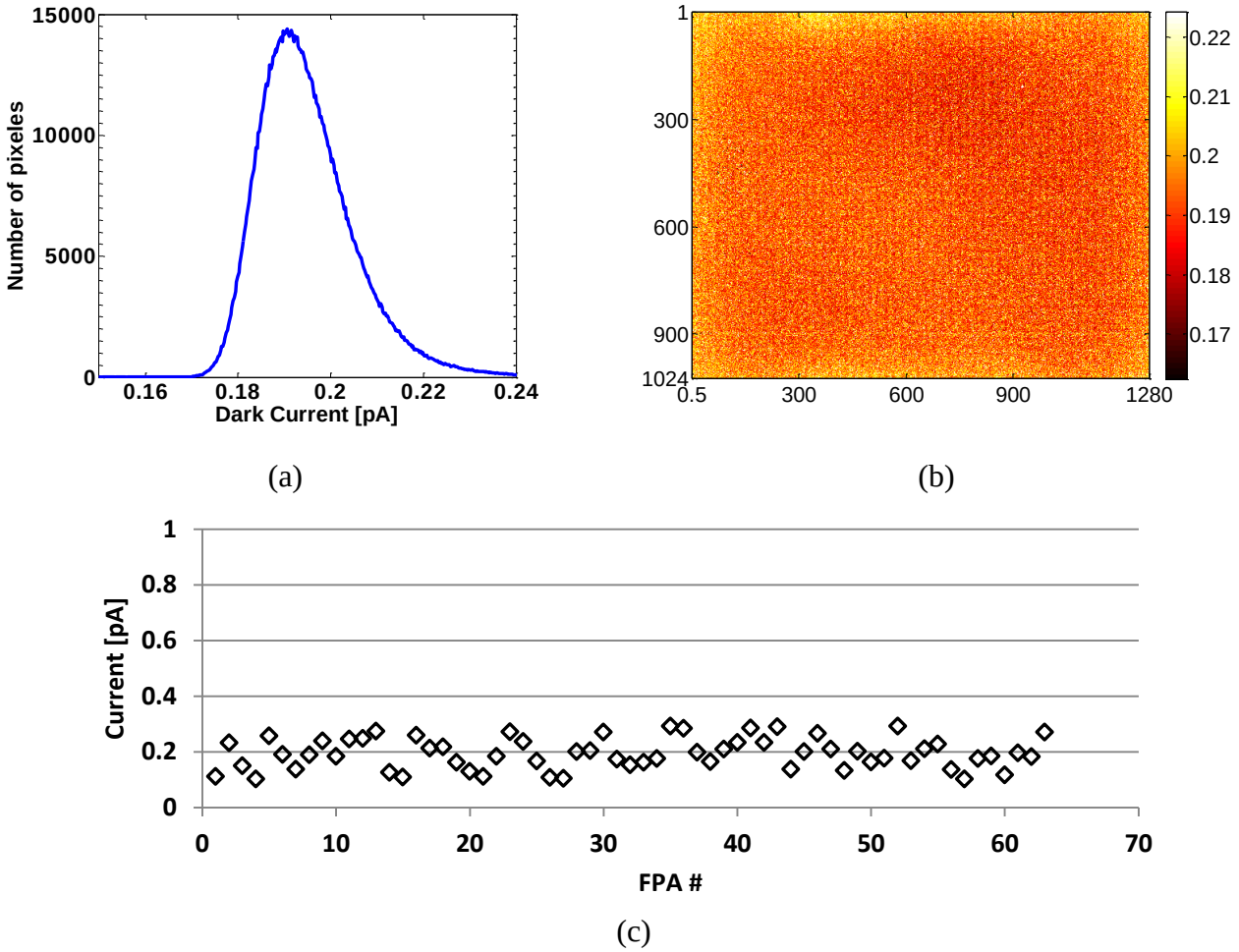


Fig. 5 Dark current (in pico-Amperes) measured in XBn-InAsSb Blackbird-1280 FPA at 150 K and plotted as (a) pixel distribution in the array, (b) 2D map, (c) mean values of 63 FPAs from SCD's production line

5. ELECTRO-OPTICAL PERFORMANCE

In addition to the values of the dark current and QE, as presented above, there are several other key properties that characterize the sensitivity, uniformity, and linear response of an FPA detector. Unlike the dark current, QE, or cross talk, which depend essentially on the photo-sensitive devices alone, other properties can be traced back to a combination of the sensing device and ROIC. One such property is the sensitivity of the detector, which is related to the temporal noise of a pixel, and is normally defined by the Noise Equivalent Temperature Difference (NETD). Another two are the

uniformity and the linear response of the pixels, which both contribute to the residual spatial fixed pattern noise that remains after performing a linear Non Uniformity Correction (NUC). Their effect can be expressed by the Residual Non Uniformity (RNU), which is the standard deviation of all pixels with respect to the NUC linear calibration, at a particular well-fill. The NETD at a given frame rate and F/# is one of the critical parameters for the evaluation of an IR detector, and is therefore an important figure of merit. It is a measure of the detector's ability to register a temperature difference that creates a signal larger than the noise. Due to the low readout noise and low dark current in Blackbird FPAs, the NETD is background limited (BLIP) even at low integration capacitor well-fill (low signal).

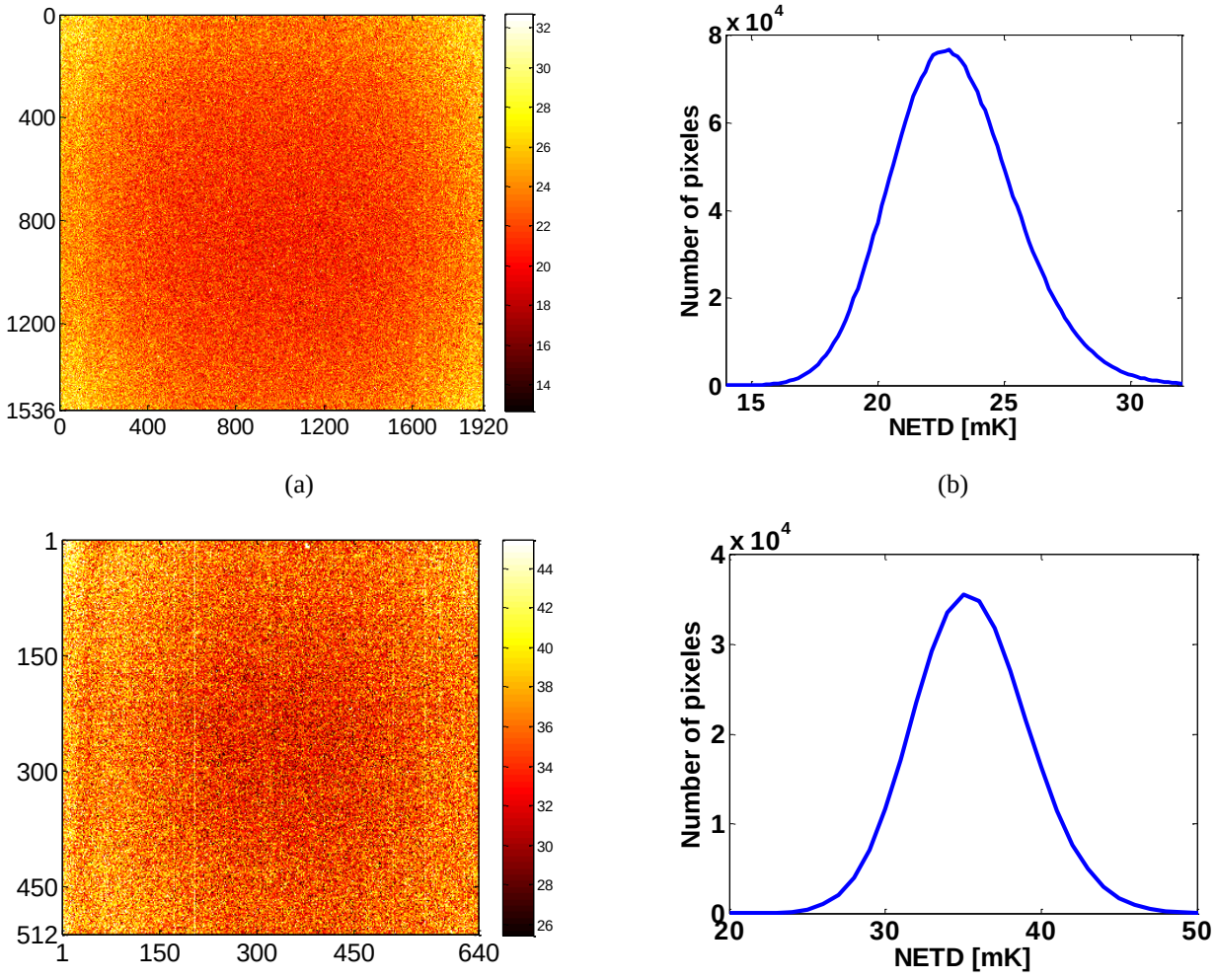


Fig. 6 (a) Typical NETD image of Blackbird-1920 with InSb diode at F/4, 77 K and 70% well-fill. The color scale is in mK (b) NETD histogram of the data in figure 6(a). (c) Typical NETD image of Blackbird-640 XBn-InAsSb at F/3, 150 K and 50% well-fill. (d) NETD histogram of the data in figure 6(c). The average NETD is 22.5 mK and the standard deviation is 2.5 mK for the Blackbird-1920 with InSb diode and it is 36 mK and the standard deviation is 4 mK for the Blackbird-640 XBn-InAsSb.

The NETD is usually calculated at an averaged signal, corresponding to a median well fill. In Figure 6(a), a typical map of the NETD (per pixel) at 70% well-fill is presented for the 2.0Me- Blackbird integration mode and a 1920×1536 InSb FPA and In Figure 6(c), a typical map of the NETD at 50% well-fill is presented for the 2.0Me- Blackbird integration mode and a 640×512 XBn-InAsSb FPA . As can be seen in the image, there are no spatial features in the temporal noise, indicating no additional noise mechanisms aside from shot noise. In Figures 6(b) and 6(d), a smooth Gaussian-like histogram is shown for the distribution of the NETD over all FPA pixels.

SCD is now able to manufacture Blackbird FPAs with good reproducibility and a low number of defective elements, corresponding to a pixel operability higher than 99.5%. In Figure 7 it is shown that the NETD does not vary significantly from FPA to FPA, for both 1920×1536 and 1280×1024 formats, and for InSb and XBn arrays. For comparison the NETD here is measured at 60% Well-Fill in all cases.

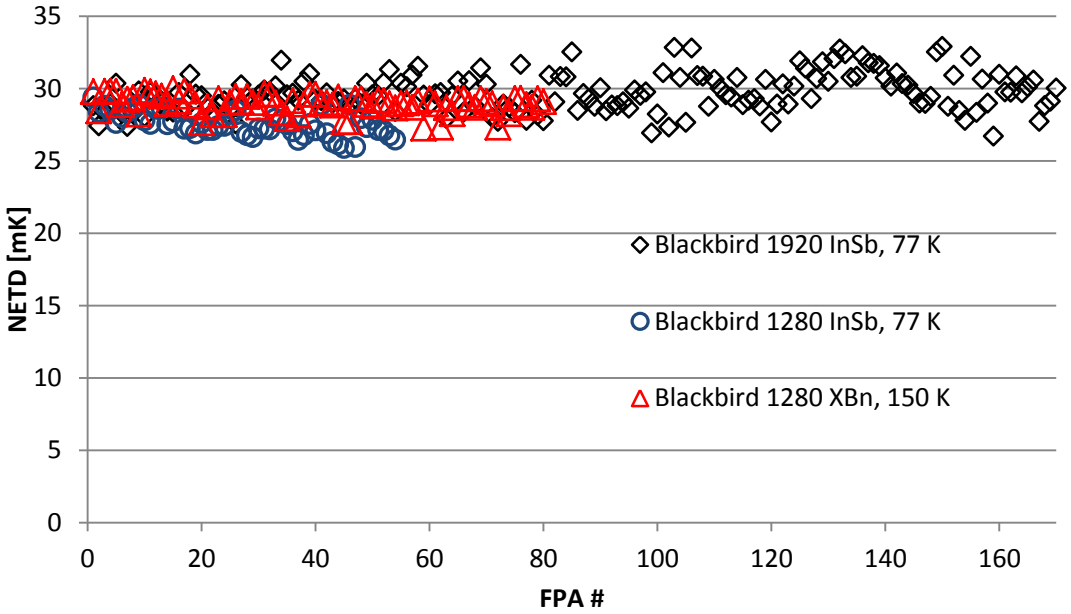
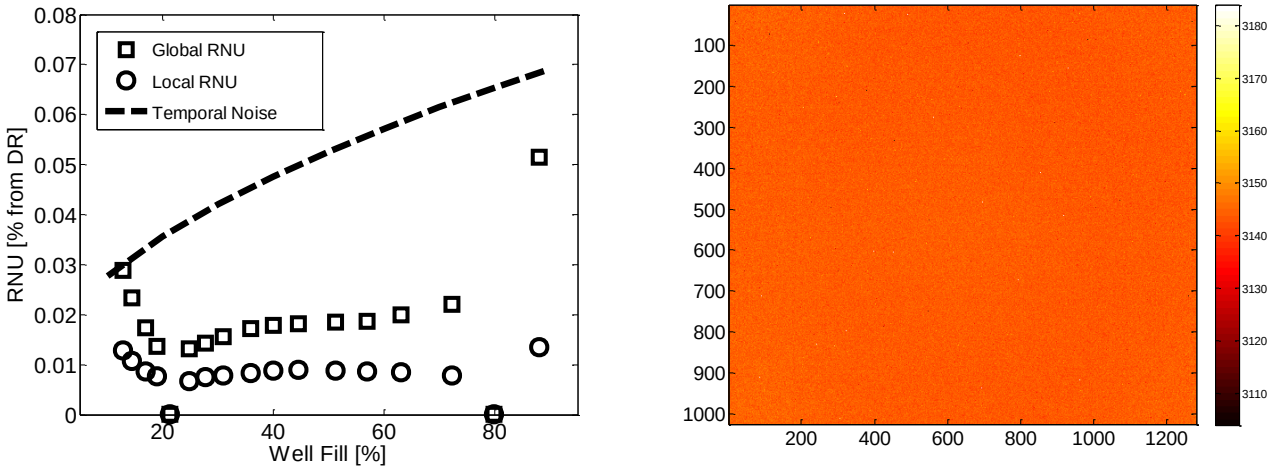


Fig. 7 Averaged NETD values for various FPAs from SCD's Blackbird production line

The detector uniformity is typically evaluated from an analysis of the corrected image after a NUC procedure. The RNU is the spatial standard deviation of the corrected image. Here we present the RNU measured after a standard linear 2-point NUC, with calibration points at well-fill levels of 20% and 80% from the full dynamic range. The RNU is inspected for various signal levels corresponding to a wide range of well fills that cover almost the entire DR. When calculated globally for all pixels in the array the RNU is affected by both low and high spatial pattern frequencies in the recorded image, and it is thus termed global RNU. High frequency patterns usually originate from spatial inhomogeneity across nearby FPA pixels. They are related to variations of the parameters of the individual pixels in both the ROIC and photo-sensitive device arrays, and can often be traced back to the fabrication processes of the two. This type of non-uniformity has white noise characteristics, is local in nature, and determines the ability of the detector to distinguish targets from their close environment. It is therefore useful to discriminate the high frequency spatial patterns from the

low frequency patterns. To that end, we define the local RNU as the standard deviation (STD) calculated over the 15×15 neighbors around a given pixel in the corrected image and averaged for all pixels. As can be seen in Figure 8(a) for an XBn FPA, it is lower than the global RNU since low frequency patterns are filtered out. The global (local) RNU of the Blackbird FPA is less than 0.025% (0.015%) STD/DR for a wide range of signal well fills. A comparison of the spatial noise (RNU) and temporal noise (related to NETD and plotted as a dashed line) indicates the high uniformity of the array. For a good BLIP detector, the spatial noise should always be significantly lower than the temporal noise, as indeed occurs in Figure 8(a). Figure 8(b) shows a corrected image of a uniform target at 50% well-fill registered by the XBn FPA in Figure 8(a). The color scale is in digital levels, and shows excellent uniformity at the individual bit-level, consistent with the very low values of RNU in Figure 8(a). Figure 8(b) also demonstrates the very high pixel operability of the FPA, since no Bad Pixel Replacement (BPR) routine was applied.

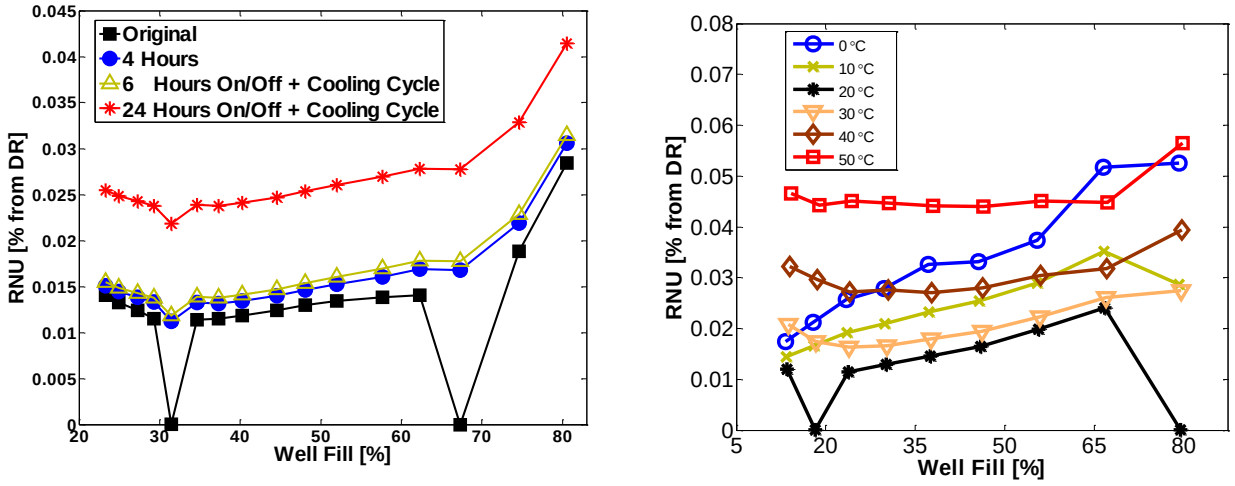


(a) (b)

Fig. 8 (a) Typical RNU from the Blackbird-1280 XBn FPA at F/4, 150 K as a function of well-fill. The signal is varied by changing the black-body target temperature at constant integration time. (b) An image of a uniform target at 50% well-fill after 2-point NUC. The color scale is in digital levels

Another important parameter of image quality is related to the stability of the RNU (and so the validity of the NUC tables) over time, from one operation to another and when the ambient temperature is changed. In order to test the RNU stability over time and operation cycles, we have used a standard F/3 Blackbird InSb Integrated Detector Cooler Assembly (IDCA) to obtain a few sets of measurements. The second set was measured four hours after the first while the detector remained cooled and powered, while the third set was measured two hours later, in this case after the ROIC power supply had been turned off and then on again. The last set was measured one day later during which time the FPA reached room temperature, before being cooled back down to 77K prior to the measurement. The Local RNU of all the data sets is calculated as described above from the Gain and Offset tables of the "original" first set. The results are

presented in Figure 9(a) for a Blackbird 1280 InSb detector although Blackbird XBn detectors show similar results. Similarly the Global RNU stability for various ambient temperatures is tested in an environmental chamber. Here the "original" gain and offset are measured at 20°C and the detector remains cooled and powered during the whole set of measurements which are presented in Figure 9(b). It is evident that the RNU is very stable during an operation cycle (constant temperature and power) and after an On/Off procedure, while it degrades moderately after a cooling cycle. It is also relatively insensitive to environmental conditions, in this case between 0 - 50°C.

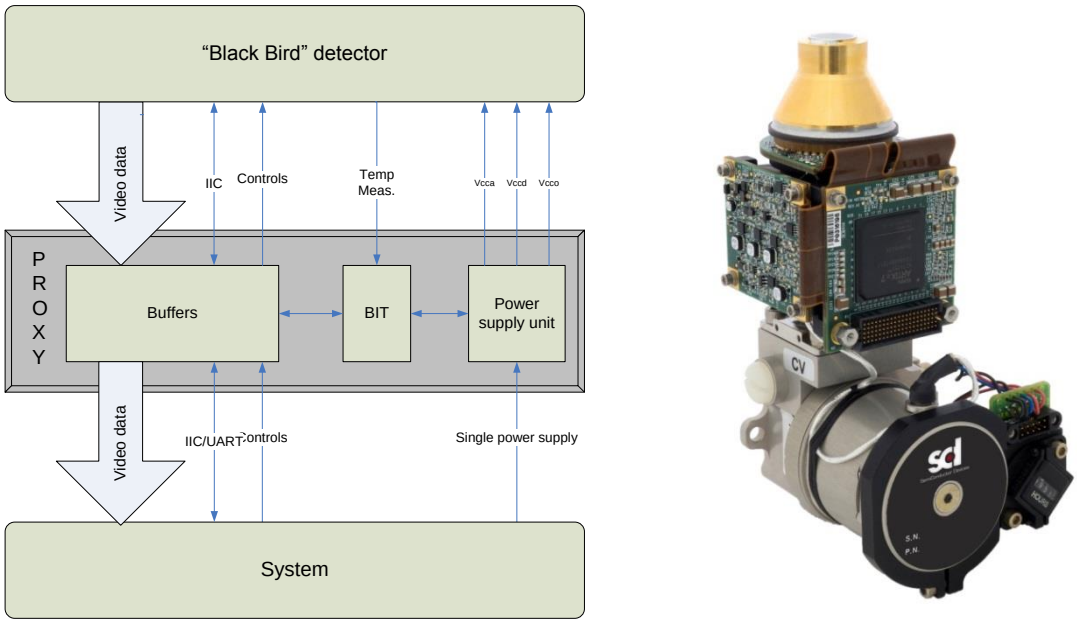


(a) (b)

Fig. 9 Stability of NUC for a Blackbird-1280 InSb IDCA at F/3 and cutoff wavelength filter of 4.2μm. The RNU is plotted as a function of well-fill. (a) Local RNU at different operation times but corrected with the same initial tables of NUC coefficients. (b) Global RNU measured at different ambient temperatures after performance of NUC at 20ºC

6. DETECTOR CONFIGURATION

The FPA is packaged in a Dewar which is integrated with a cryo-cooler and an electronics proximity board suitable to its format. A schematic of a typical proximity electronics layout and an image of the IDCA itself, is shown in Figures 10(a) and 10(b), respectively. SCD has adapted its well established rigid Dewar technology from Blackbird's predecessors to fit the current set of detectors.



(a) (b)

Fig. 10 (a) Scheme of the proximity electronics. (b) The Blackbird-1280 IDCA

The rugged Dewar envelope has supporting strings which are connected to the cold finger. The structure and the geometry were optimized to give a high natural frequency. This results in a sub pixel lateral movement of the FPA when the complete IDCA is subjected to rough vibrations in the frequency range 5-2000Hz. The IDCA makes a compact MWIR detector that can withstand harsh environmental conditions such as high ambient temperatures of up to 71 ºC. The design and fabrication of the Dewar minimizes the heat load and stray light on the FPA.

A demonstration of the high performance of the Dewar comes from a measurement of the Excess Current, namely the sum of the dark current and the current due to stray light. In Figure 11, the excess current in Blackbird InSb 1280 IDCA with F#3.6 is measured in front of a uniform extended blackbody and plotted as a function of the ambient temperature. By comparing to the optical model (dash line) we conclude that these results correspond to a very low effective emissivity of 2.6% from the surrounding 2π window housing of the Dewar.

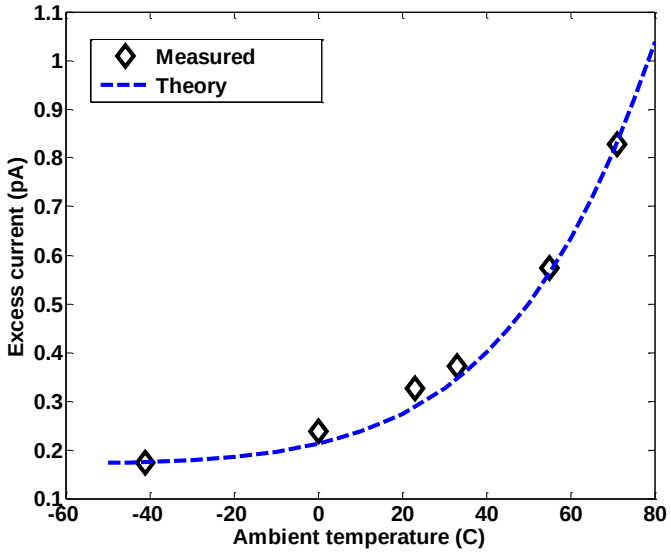
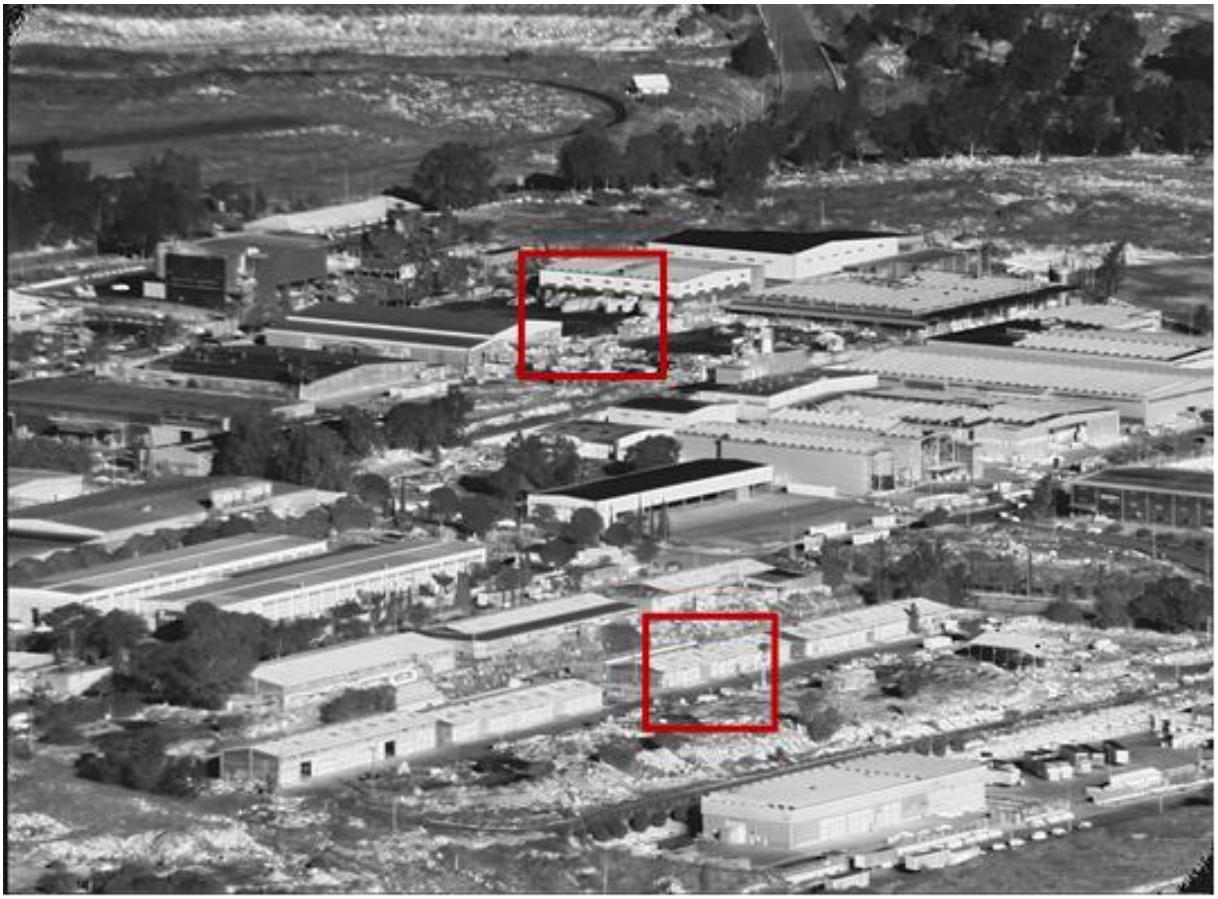


Fig. 11 Excess Current as a function of ambient temperature as measured (open diamonds) and simulated (dash line) in a Blackbird InSb1280 IDCA with F#3.6 cold stop

The proximity electronics board has a single power supply and a standard video output and communication that enable a fast and easy integration of the detector into the system. The Blackbird family is designed to easily replace the previous generation SCD IDCAs (Hercules and Pelican-D) with compatible electronics and mechanical interfacing, including the cryo-coolers.

7. SUMMARY

In conclusion, Blackbird is a diverse family of MWIR detectors with a range of formats based on both cooled InSb and HOT XBn technologies. The Blackbird detectors exhibit high electro-optical performance and feature many qualities beneficial to the system, making them good candidates for a wide spectrum of modern IR system applications. Moreover, the detector's standard mechanical and electrical interface allows easy and fast integration into existing systems to improve performance and to reduce both system Size, Weight, and Power and also system Cost. Finally, an example of the kind of wide field of view and high resolution image that can be obtained with the largest format 1920×1536 Blackbird detector, in this case an InSb sensor, is shown in Figure 12.



(a)



(b)



(c)

Fig. 12 (a) Image from the Blackbird-1920 InSb detector with 300 mm lens from 5 km away. (b) & (c) Digital zoom of selected regions in the image indicated by the red rectangles in (a)

REFERENCES

[1] R.G. Driggers, R. Vollmerhausen, J.P Reynolds, J. Fanning, and G.C. Holst, "*How low should you go?*" Optical Engineering **51**, 063202: 0-6 (2012).

[2] P.C. Klipstein "*Depletionless Photodiode with Suppressed Dark Current…*", US Patent 7,795,640 (2 July 2003)

[3] L. Shkedy, M. Brumer, P. Klipstein, M. Nitzani, E. Avnon, Y. Kodriano, I. Lukomsky, and I. Shtrichman, "*Development of 10µm pitch XBn detector for low SWaP MWIR applications* " Proc. SPIE **9819**, 9819-1D (2016).

[4] O. Nesher and P. C. Klipstein, "*High performance IR detectors at SCD present and future*", Opto-Electronics Review **14**, 61 (2005)

[5] L. Shkedy, T. Markovitz, Z. Calahorra, I. Hirsh, and I. Shtrichman, " *Megapixel digital InSb detector for infrared imaging*". Optical Engineering **50**, 061008 (2011)

[6] E. Ilan, N. Shiloah, S. Elkind, R. Dobromislin, W. Freiman, A. Zviagintsev, I. Nevo, O. Cohen, F. Khinich, A. Adin, R. Talmor, and Y. Milstain," *A 3Mpixel ROIC with 10um pixel pitch and 120 Hz frame rate digital output* " Proc. SPIE **8659**, 8659-0A (2013)

[7] G. Gershon, A. Albo, M. Eylon, O. Cohen, Z. Calahorra , M. Brumer, M. Nitzani, E. Avnon, Y. Aghion, I. Kogan, E. Ilan, and L. Shkedy " *3 Mega-pixel InSb Detector with 10 µm Pitch*" Proc. SPIE **8704**, 8704-38 (2013)

[8] G. Gershon, A. Albo, M. Eylon, O. Cohen, Z. Calahorra, M. Brumer, M. Nitzani, E. Avnon, Y. Aghion, I. Kogan, E. Ilan, A. Tuito, M. Ben Ezra and L. Shkedy " *Large Format InSb Infrared Detector with 10 µm Pixels*" Proc. OPTRO-2014-2931891 (2014)

[9] P.C. Klipstein, O. Klin, S. Grossman, N. Snapi, B. Yaakobovitz, M. Brumer, I. Lukomsky, D. Aronov, M. Yassen, B. Yofis, A. Glozman, T. Fishman, E. Berkowicz, O. Magen, I. Shtrichman, and E. Weiss "*XBn Barrier Detectors for High Operating Temperatures*", Proc. SPIE **7608**, 7608-1V (2010)

[10] I. Shtrichman, D. Aronov, M. ben Ezra, I. Barkai, E. Berkowicz, M. Brumer, R. Fraenkel, A. Glozman, S. Grossman, E. Jacobsohn, O. Klin, P.C. Klipstein, I. Lukomsky, L. Shkedy, N. Snapi, M. Yassen and E. Weiss " *High Operating Temperature epi-InSb and XBn-InAsSb photodetectors* " Proc. SPIE **8353**, 8353-2Y (2012)

[11] P.C. Klipstein, Y. Gross, D. Aronov, M. ben Ezra, E. Berkowicz, Y. Cohen, R. Frenkel, A. Glozman, S. Grossman, O. Klin, I. Lukomsky, T. Markowitz, L. Shkedy, I. Shtrichman, N. Snapi, A. Tuito, M. Yassen, and E. Weiss (2012) " *Low SWaP MWIR detector based on XBn Focal Plane Array* ", Proc. SPIE **8704**, 8704-1S.